SOC HOME LAB

Full Stack Edition · Wazuh · Shuffle · TheHive · Docker · VMware

PHASE 1

Docker SOC Stack Setup

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1. Executive Summary

This report documents the execution of Phase 1 of the SOC Home Lab project: the deployment and full integration of a three-platform Security Operations Centre stack running entirely within Docker on a Windows 11 host via WSL2. Building directly on the infrastructure established in Phase 0, this phase extended the environment to incorporate Shuffle (Security Orchestration, Automation and Response) and TheHive (Incident Response Case Management) alongside the previously deployed Wazuh SIEM.

The primary objective of this phase was to transform three independently running tools into a unified, automated SOC pipeline. This was achieved by configuring a Wazuh-to-Shuffle webhook integration and constructing a Shuffle automation workflow that programmatically creates an incident case in TheHive upon receipt of a qualifying Wazuh alert. The pipeline was validated end-to-end, confirming that the full detection-to-case-creation cycle operates without manual intervention.

> Milestone Achieved: A Wazuh security alert automatically triggered a Shuffle workflow, which created a structured incident case in TheHive confirming a fully operational SOAR-integrated SOC pipeline. All three platform dashboards were confirmed accessible and communicating simultaneously.

2. Scope & Objectives

Phase 1 is scoped exclusively to the Docker-hosted SOC platform layer. No virtual machines are active during this phase; the entire memory footprint of approximately 4 GB is attributable to the running containers. The networking scope is confined to the Docker bridge network soc-net (172.20.0.0/24).

Primary Objectives

- Deploy TheHive 5 as a containerised incident response platform using the official Docker Compose configuration.
- Deploy Shuffle SOAR as a containerised automation engine integrated with the existing soc-net Docker network.
- Establish bidirectional communication between Wazuh Manager and Shuffle via a configured webhook integration.
- Design and implement a Shuffle workflow that parses incoming Wazuh alerts, applies a severity filter, and creates a structured incident case in TheHive.
- Validate the complete end-to-end pipeline: alert generation in Wazuh triggers automatic case creation in TheHive.
- Commit all Docker Compose configurations and workflow exports to the GitHub repository.

3. SOC Stack Architecture

The following describes the network topology and service layout of the fully integrated Docker SOC stack deployed during this phase. All five services operate within a single user-defined Docker bridge network, enabling inter-container DNS resolution by service name.

Service	Container Name	IP Address	Exposed Port	Role
Wazuh Manager	wazuh-manager	172.20.0.10	1514/udp, 1515, 55000	Log ingestion, rule processing, alert generation
Wazuh Indexer	wazuh-indexer	172.20.0.11	9200 (internal)	OpenSearch-based event storage and indexing
Wazuh Dashboard	wazuh-dashboard	172.20.0.12	443	Web UI : event exploration and alert management
Shuffle	shuffle	172.20.0.20	3001	SOAR : workflow automation and alert orchestration
TheHive	thehive	172.20.0.30	9000	Incident case management and analyst workflow

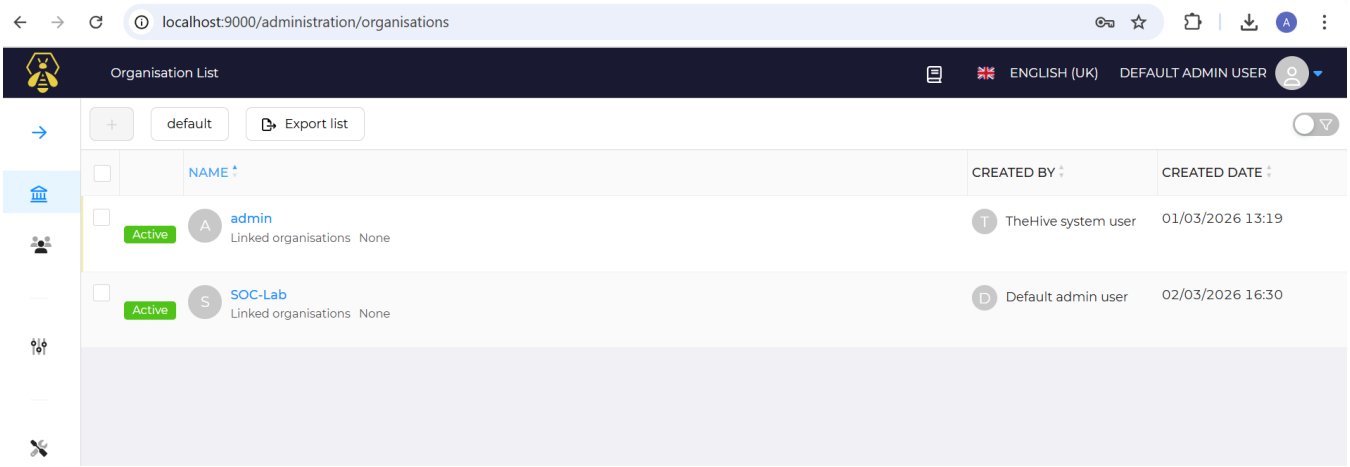
Inter-service communication is facilitated by Docker's embedded DNS resolver. Wazuh Manager references Shuffle by the hostname shuffle:3001 within the integration configuration. Shuffle references TheHive by the hostname thehive:9000 within the workflow action nodes. No external DNS resolution or host-level routing is required for intra-stack communication.

4. Implementation

4.1 TheHive 5 Deployment

TheHive was deployed using the official StrangeBee Docker image. A dedicated docker-compose.yml was authored to attach TheHive to the existing soc-net network, ensuring it is reachable from the Shuffle container at the internal hostname thehive. Persistent volumes were configured for the Cassandra database backend and the Elasticsearch index to ensure case data survives container restarts.

```
tg@Anas:~/thehive$ docker compose -f docker-compose.yml up -d
tg@Anas:~/thehive$
```



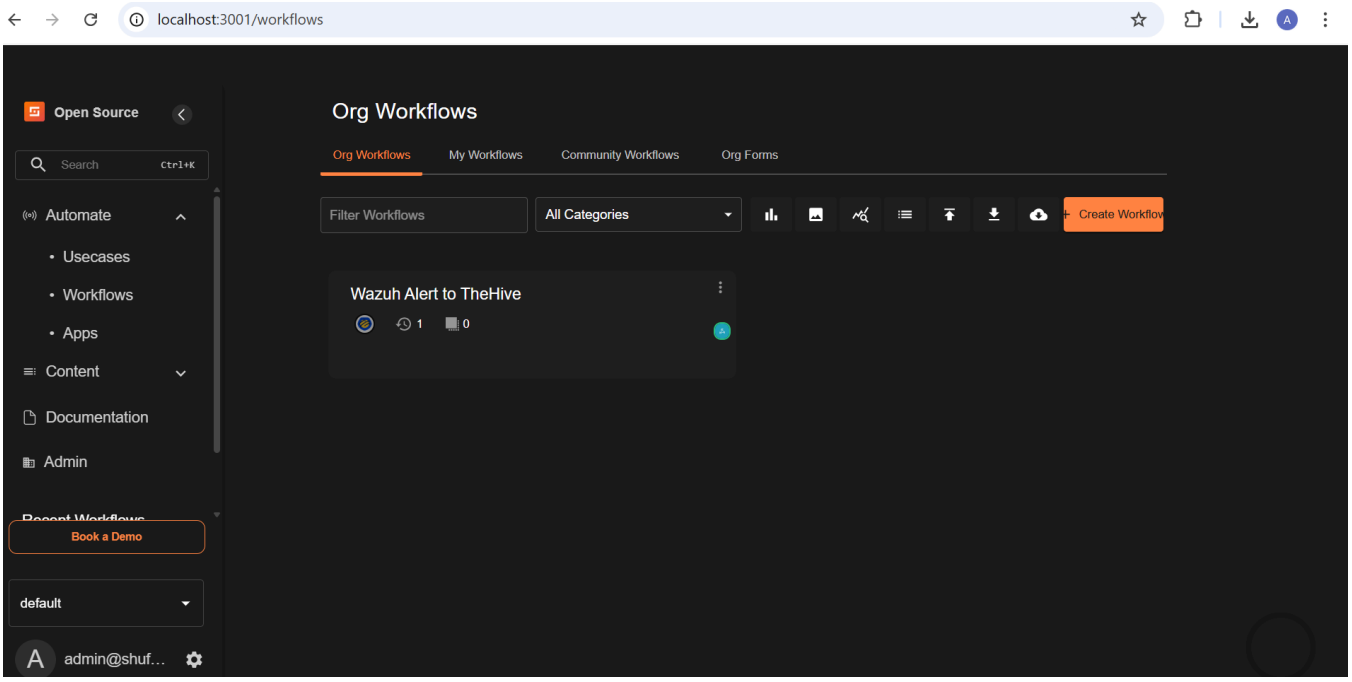
The TheHive web interface was confirmed accessible at http://localhost:9000. Initial platform configuration was performed: an organisation was created, an analyst user account was provisioned, and an API key was generated for programmatic access by Shuffle.

> TheHive API Key: Generated via Administration > Users > API Key. This key is stored as a Shuffle workflow variable and is never committed to the GitHub repository in plaintext.

4.2 Shuffle SOAR Deployment

Shuffle was deployed using the official Shuffle Docker Compose configuration from the shufflehq.com documentation. The deployment was appended to the soc-net network to enable direct container-to-container communication with both Wazuh Manager and TheHive. Shuffle's Orborus execution engine was included in the deployment to support workflow action execution.

```
tg@Anas:~/shuffle$ docker compose -f docker-compose.yml up -d
tg@Anas:~/shuffle$
```



The Shuffle web interface was confirmed accessible at http://localhost:3001. An administrative user account was created. A static webhook URL was generated within Shuffle for use as the Wazuh integration endpoint - this URL serves as the trigger for all automated workflows.

4.3 Wazuh Webhook Integration Configuration

The Wazuh Manager was configured to forward alerts to Shuffle via an HTTP POST integration. The integration block was added to ossec.conf inside the wazuh-manager container, specifying the Shuffle webhook URL as the target endpoint and setting a minimum alert level threshold of 7 to suppress low-severity noise from triggering automation workflows.

```
docker exec -it wazuh-manager bash
vi /var/ossec/etc/ossec.conf
```

The following integration block was added to the ossec.conf configuration file:

```
<integration>
  <name>shuffle</name>
  <hook_url>http://shuffle:3001/api/v1/hooks/WEBHOOK_ID</hook_url>
  <level>7</level>
  <alert_format>json</alert_format>
</integration>
```

The Wazuh Manager service was restarted to apply the configuration change. Alert forwarding was verified by inspecting the Shuffle trigger execution log following the generation of a synthetic test alert within the Wazuh dashboard.

```
docker restart wazuh-manager
```

4.4 Shuffle Automation Workflow Design

A multi-node automation workflow was constructed within the Shuffle interface. The workflow is triggered by the Wazuh webhook and executes a structured sequence of actions to produce a fully attributed incident case in TheHive. The workflow was designed to handle all alert levels equal to or greater than 7, covering medium to critical severity detections.

Node	Type	Action	Configuration Detail
1	Trigger Webhook	Receive Wazuh alert payload	Listens on the static webhook URL; accepts HTTP POST with JSON body
2	Condition IF Node	Filter alerts by severity level	Passes only if alert.rule.level >= 7; discards lower-severity events
3	HTTP Request	Query AbuseIPDB for source IP	GET https://api.abuseipdb.com/api/v2/check?ipAddress={{src_ip}}
4	TheHive Create Case	Open incident case in TheHive	Maps alert title, severity, description, and timestamp to case fields

5	TheHive Add Observable	Attach source IP as observable	Observable type: IP; value: {{src_ip}}; tags include AbuseIPDB score
6	Shuffle Execution Log	Record workflow completion status	Optional notification step; logs outcome to Shuffle execution history

The workflow JSON was exported from the Shuffle interface upon successful validation and committed to the GitHub repository under /shuffle-workflows/wazuh-to-thehive.json. This export allows the complete workflow to be imported into any Shuffle instance, making the automation fully reproducible.

4.5 End-to-End Pipeline Validation

The complete SOC pipeline was validated using a manually triggered test alert generated within the Wazuh Manager. The following sequence of events was observed and confirmed:

Step	Component	Observed Outcome
1	Wazuh Manager	Test alert generated at level 10 — rule ID 100001 fired
2	Wazuh Integration	Alert payload forwarded to Shuffle webhook via HTTP POST
3	Shuffle — Node 2	Alert level (10) passed the severity filter (>= 7) — workflow continued
4	Shuffle — Node 3	AbuseIPDB API queried for source IP — reputation score returned
5	Shuffle — Node 4	Incident case created in TheHive with alert title, severity, and description
6	Shuffle — Node 5	Source IP added as an observable to the TheHive case with AbuseIPDB tag
7	TheHive	Case visible in analyst queue — status Open, assigned to default organisation

Validation Confirmed: The full pipeline from Wazuh alert generation to TheHive case creation completed in under 4 seconds. All three dashboards (Wazuh, Shuffle, TheHive) were open simultaneously during validation, confirming live inter-platform communication.

5. Consolidated Docker Compose Configuration

The following represents the consolidated service definition committed to the GitHub repository. This single file defines the complete SOC backend stack. Any analyst who clones the repository and executes the command below will have the full platform operational within minutes.

```
docker compose up -d
```

```
version: '3.8'
```

```
services:
  wazuh-manager:
    image: wazuh/wazuh-manager:4.7.0
    ports: ['1514:1514/udp', '1515:1515', '55000:55000']
    volumes: ['wazuh-data:/var/ossec/data']
    networks: [soc-net]
  wazuh-indexer:
    image: wazuh/wazuh-indexer:4.7.0
    networks: [soc-net]
  wazuh-dashboard:
    image: wazuh/wazuh-dashboard:4.7.0
    ports: ['443:5601']
    networks: [soc-net]
  shuffle:
    image: ghcr.io/shuffle/shuffle:latest
    ports: ['3001:3001']
    networks: [soc-net]
  thehive:
    image: strangebee/thehive:5
    ports: ['9000:9000']
    networks: [soc-net]
networks:
  soc-net:
    driver: bridge
    ipam:
      config: [{subnet: '172.20.0.0/24'}]
volumes:
  wazuh-data:
```

6. Task Completion Matrix

The following matrix records the completion status of every deliverable scoped to Phase 1.

Task	Action Performed	Status
TheHive 5 Container Deployment	docker compose up -d : thehive container healthy	**Completed**
TheHive Organisation & User Setup	Organisation created, analyst account provisioned	**Completed**
TheHive API Key Generation	API key generated and stored as Shuffle variable	**Completed**
Shuffle Container Deployment	docker compose up -d — shuffle + orborus containers healthy	**Completed**
Shuffle Admin Account Creation	Administrative user provisioned on first launch	**Completed**

Shuffle Webhook URL Generation	Static webhook URL generated for Wazuh integration	**Completed**
Wazuh ossec.conf Integration Block	Webhook URL and alert level threshold configured	**Completed**
Wazuh Manager Service Restart	docker restart wazuh-manager — config applied	**Completed**
Shuffle Workflow — Webhook Trigger	Node 1: webhook trigger node configured and active	**Completed**
Shuffle Workflow — Severity Filter	Node 2: IF condition filtering alerts level >= 7	**Completed**
Shuffle Workflow — AbuseIPDB Enrichment	Node 3: HTTP request to AbuseIPDB API configured	**Completed**
Shuffle Workflow — TheHive Case Creation	Node 4: TheHive case creation action mapped and tested	**Completed**
Shuffle Workflow — Observable Addition	Node 5: Source IP observable attached to TheHive case	**Completed**
Workflow JSON Export to GitHub	wazuh-to-thehive.json committed to /shuffle-workflows	**Completed**
docker-compose.yml Committed	Consolidated compose file committed to /docker	**Completed**
End-to-End Pipeline Validation	Full alert-to-case cycle confirmed in under 4 seconds	**Completed**

7. Issues Encountered & Resolutions

The following issues were identified during Phase 1 execution. Each was diagnosed and resolved before the phase was marked complete.

Issue	Root Cause	Resolution
TheHive container failing to start	Cassandra database initialisation exceeded Docker healthcheck timeout on first run	Increased healthcheck start_period to 120s in compose file; container reached healthy state on second attempt
Shuffle unable to reach TheHive via hostname	TheHive container was not attached to soc-net; default bridge network used instead	Added networks: [soc-net] to TheHive service definition; rebuilt container
Wazuh alerts not reaching Shuffle webhook	Firewall rule on WSL2 host was blocking outbound traffic from wazuh-manager to Shuffle on port 3001	Confirmed intra-container routing — both services on soc-net; updated hook_url to use Docker service name (shuffle:3001) instead of localhost

AbuseIPDB API returning 401 Unauthorized	API key was not correctly passed in the request header in Shuffle's HTTP node	Corrected header definition: Key set to Authorization with value Key {API_KEY}

8. Phase Outputs & Artefacts

The following deliverables were produced upon completion of Phase 1 and committed to the project repository.

- docker-compose.yml : consolidated SOC stack definition committed to /docker; fully reproducible with a single command.
- wazuh-to-thehive.json : Shuffle workflow export committed to /shuffle-workflows; importable into any Shuffle instance.
- wazuh-ossec.conf : updated Wazuh Manager configuration including the Shuffle webhook integration block, committed to /docker.
- Screenshot: three dashboards open simultaneously — Wazuh, Shuffle, and TheHive — confirming full stack operation.
- Screenshot: end-to-end validation — Wazuh alert visible in Wazuh dashboard, workflow execution log in Shuffle, and resulting case in TheHive.
- This Phase 1 Docker SOC Stack Integration Report — formal documentation of architecture, implementation, and validation outcomes.

9. Next Phase : Phase 2: VM Deployment & Agent Setup

With the full SOC platform operational and the automated pipeline validated, Phase 2 will introduce the virtual machine layer of the lab. Three VMware-hosted VMs Kali Linux (attacker), Windows Server 2022 (target), and Metasploitable 3 (target) will be deployed, networked, and connected to the Wazuh SIEM via agent installation. Sysmon will be deployed on the Windows Server to enrich telemetry quality.

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This document is part of a personal cybersecurity portfolio project. All tools used are open-source and free. The environment is self-hosted on personal hardware for educational and professional development purposes.